CHAPTER 5

THE TECHNOLOGY OF LIFE

LIFE, THE UNIVERSE'S MOST ANCIENT technology, is at least 3.7 billion years old. Across these eons life evolved in a glacial, self-governing, and unguided process. Then, in just the past few decades, the tiniest sliver of evolutionary time, one of life's products, humans, changed everything. Biology's mysteries began to unravel, and biology itself became an engineering tool. The story of life had been rewritten in an instant; the meandering hand of evolution suddenly supercharged, given direction. Changes that once unfolded blindly and on geological time now careen forward at an exponential pace. Alongside AI, this is the most important transformation of our lifetimes.

Living systems self-assemble and self-heal; they're energy-harnessing architectures that can replicate, survive, and flourish in a vast range of environments, all at a breathtaking level of sophistication, atomic precision, and information processing. Just as everything from the steam engine to the microprocessor was driven by an intense dialogue between physics and engineering, so the coming decades will be defined by a convergence of biology and engineering. Like AI, synthetic biology is on a sharp trajectory of falling costs and rising capabilities.

At the center of this wave sits the realization that DNA *is* information, a biologically evolved encoding and storage system. Over recent decades we have come to understand enough about this information transmission system that we can now intervene to alter its encoding and direct its course. As a result, food, medicine, materials, manufacturing processes, and consumer goods will all be transformed and reimagined. So will humans themselves.

DNA SCISSORS: THE CRISPR REVOLUTION

Genetic engineering sounds modern, but it's actually one of humankind's oldest technologies. Much of civilization would have been impossible without selective breeding—the insistent process of refining crops and animals to select for more desirable traits. Steadily, over centuries and millennia, humans bred for traits that would be most useful, producing friendly dogs, dairy cattle, domesticated chickens, wheat, corn, and so on.

Modern bioengineering began in the 1970s, building on a growing understanding of heredity and genetics that had started in the nineteenth century. Extending the work of Rosalind Franklin and Maurice Wilkins, James Watson and Francis Crick discovered the structure of DNA, the molecule encoding instructions for producing an organism, in the 1950s. Then, working on bacteria in 1973, Stanley N. Cohen and Herbert W. Boyer found ways of transplanting genetic material from one organism into another, showing how they could successfully introduce DNA from a frog into a bacterium. The age of genetic engineering had arrived.

This research led Boyer to found one of the world's first biotech companies, Genentech, in 1976. Its mission was to manipulate the genes of microorganisms to produce medicines and treatments, and within a year it had developed a proof of concept, using engineered *E. coli* bacteria to produce the hormone somatostatin.

Despite some notable achievements, initial progress in the field was slow, because genetic engineering was a costly, difficult process prone to failure. Over the last twenty or so years, however, that has changed. Genetic engineering has gotten much cheaper and much easier. (Sound familiar?) One catalyst was the Human Genome Project. This was a thirteen-year, multibillion-dollar endeavor that gathered together thousands of scientists from across the world, in private and public institutions, with a single goal: unlocking the three billion letters of genetic information making up the human genome. Genome sequencing like this turns biological information,

DNA, into raw text: information humans can read and use. Complex chemical structure is rendered into a sequence of its four defining bases—A, T, C, and G.

For the first time, the Human Genome Project aimed to make the full genetic map of human beings legible. When it was announced in 1988, some thought it was impossible, doomed. But the project eventually proved the doubters wrong. By 2003, it was announced at a White House ceremony that 92 percent of the human genome had been sequenced and the code of life was now laid bare. It was a landmark achievement, and though it has taken time to start reaching its full potential, in hindsight, it's clear that the Human Genome Project really did mark the beginning of a revolution.

While Moore's law justifiably attracts considerable attention, less well known is what *The Economist* calls the Carlson curve: the epic collapse in costs for sequencing DNA. Thanks to ever-improving techniques, the cost of human genome sequencing fell from $1 billion in 2003 to well under $1,000 by 2022. That is, the price dropped *a millionfold* in under twenty years, a thousand times faster than Moore's law. A stunning development hiding in plain sight.

Genome sequencing is now a booming business. In time it seems likely that the majority of people, plants, animals, and everything in between will have their genomes sequenced. Services like 23andMe already offer DNA profiling of individuals for a few hundred dollars.

But the power of biotech goes far beyond our ability to simply read the code; it now enables us to edit it, and write it, too. CRISPR gene editing (the acronym stands for clustered regularly interspaced short palindromic repeats) is perhaps the best-known example of how we can directly intervene in genetics. A breakthrough in 2012 led by Jennifer Doudna and Emmanuelle Charpentier meant that for the first time genes could be edited almost like text or computer code, far more easily than in the early days of genetic engineering.

CRISPR edits DNA sequences with the help of Cas9, an enzyme acting as a pair of finely tuned DNA scissors, cutting parts of a DNA strand for precise genetic editing and modification of anything ranging from a minute

bacterium to large mammals like human beings, with edits anywhere from tiny changes to significant interventions in the genome. Impacts can be enormous: editing germ-line cells that form eggs and sperm, for example, means changes will echo down through generations.

After the initial CRISPR paper was published, progress applying it was rapid; the first gene-edited plants were created within a year, the first animals—mice—even before that. CRISPR-based systems with names like Carver and PAC-MAN promise effective prophylactic ways of fighting viruses that, unlike vaccines, don't trigger an immune response, helping protect us against pandemics of the future. Fields like RNA editing are themselves opening a range of new treatments for conditions like high cholesterol and cancer. New techniques like Craspase, a CRISPR tool working with RNA and proteins rather than DNA, might allow for safer therapeutic interventions than conventional methods.

Like AI, genetic engineering is a field in blistering motion, evolving and developing by the week, a massive global concentration of talent and energy beginning to bear real fruit (in this case, literally). CRISPR use cases are multiplying, from tomatoes ultrarich in vitamin D to treatments for conditions including sickle-cell disease and beta-thalassemia (a blood disorder producing abnormal hemoglobin). In the future, it could offer treatments for COVID-19, HIV, cystic fibrosis, and even cancer. Safe, widespread gene therapies are on their way. These will create crops that are resistant to drought and disease, boost yields, and help enable the production of biofuels at scale.

Just a few decades ago biotech was expensive, complex, and slow moving, with only the most talented and well-resourced teams able to participate. Today technologies like CRISPR are simple and cheap to use; they have, in the words of the biologist Nessa Carey, "democratized biological science." Experiments that once took years are tackled by grad students in weeks. Companies like the Odin will sell you a genetic engineering kit including live frogs and crickets for $1,999, while another kit includes a mini-centrifuge, a polymerase chain reaction machine, and all the reagents and materials you need to get going.

Genetic engineering has embraced the do-it-yourself ethos that once defined digital start-ups and led to such an explosion of creativity and potential in the early days of the internet. You can now buy a benchtop DNA synthesizer (see the next section) for as little as $25,000 and use it as you wish, without restriction or oversight, at home in your bio-garage.

DNA PRINTERS: SYNTHETIC BIOLOGY COMES TO LIFE

CRISPR is only the start. Gene synthesis is the manufacture of genetic sequences, printing strands of DNA. If sequencing is reading, synthesizing is writing. And writing doesn't just involve reproducing known strands of DNA; it also enables scientists to write new strands, to engineer life itself. While the practice existed years ago, it was again slow, expensive, and difficult. A decade ago, scientists might have produced under a hundred pieces of DNA simultaneously. Now they can print millions at once, combined with a tenfold fall in price. The London DNA Foundry housed at Imperial College London claims it can create and test fifteen thousand different genetic designs in a single morning.

Companies such as DNA Script are commercializing DNA printers that train and adapt enzymes to build de novo, or completely new, molecules. This capability has given rise to the new field of synthetic biology—the ability to read, edit, and now write the code of life. Furthermore, new techniques like enzymatic synthesis are faster and even more efficient while being less prone to failure, without hazardous waste, and, of course, on a steep declining cost curve. The method is also much easier to learn, unlike highly complex older methods that require more specialized knowledge and technical skills.

A world of possibility for the creation of DNA has opened up, one in which cycles of designing, building, testing, and iterating happen at a radically accelerated pace. At-home versions of DNA synthesizers currently

have some technical limitations but are still enormously powerful, and you can bet those limitations will be overcome in the near future.

Where nature takes a long and winding path to reach extraordinarily effective results, this bio-revolution puts the power of concentrated design at the heart of these self-replicating, self-healing, and evolving processes.

This is the promise of evolution by design, tens of millions of years of history compressed and short-circuited by directed intervention. It brings together biotechnology, molecular biology, and genetics with the power of computational design tools. Put it all together and you have a platform of profoundly transformational scope. In the words of the Stanford bioengineer Drew Endy, "Biology is the ultimate distributed manufacturing platform." Synthetic biology's true promise, then, is that it will "enable people to more directly and freely make whatever they need wherever they are."

In the 1960s computer chips were still largely hand built, just as—until recently—most biotech research was still a manual process, slow, unpredictable, messy in every sense. Now semiconductor fabrication is a hyperefficient atomic-scale manufacturing process churning out some of the world's most complex products. Biotech is following a similar trajectory, only at a much earlier phase; organisms will soon be designed and produced with the precision and scale of today's computer chips and software.

In 2010 a team led by Craig Venter took a near copy of the genome of the bacterium *Mycoplasma mycoides* and transplanted it into a new cell that then replicated. It was, they argued, a new life-form, Synthia. In 2016 they created an organism with 473 genes, fewer than anything found in nature but a decisive advance from what was previously possible. Just three years later, a team at ETH Zurich created the first bacterial genome produced entirely on a computer: *Caulobacter ethensis-2.0.* While Venter's experiments had a large team and cost millions of dollars, this pioneering work was largely completed by two brothers for under $100,000. Now the global GP-write Consortium is dedicated to bringing the cost of producing and testing synthetic genomes down "1,000-fold within ten years."

Biology, meet exponential improvements.

BIOLOGICAL CREATIVITY UNLEASHED

Countless experiments are underway in the strange and emerging landscape of synthetic biology: viruses that produce batteries, proteins that purify dirty water, organs grown in vats, algae that draw down carbon from the atmosphere, plants that consume toxic waste. Some disease-spreading species like mosquitoes or invasive species like common house mice might be phased out of habitats in so-called gene drives; others brought back to life, including one esoteric project to reintroduce woolly mammoths to the tundra. No one can fully say what the consequences might be.

Medical advances are an obvious area of focus. Using a gene for light-detecting proteins taken from algae to rebuild nerve cells, scientists successfully restored limited vision to a blind man in 2021. Previously intractable conditions from sickle-cell disease to leukemia are now potentially treatable. CAR T-cell therapies engineer bespoke immune response white blood cells to attack cancers; genetic editing looks set to cure hereditary heart conditions.

Thanks to lifesaving treatments like vaccines, we are already accustomed to the idea of intervening in our biology to help us fight disease. The field of systems biology aims to understand the "larger picture" of a cell, tissue, or organism by using bioinformatics and computational biology to see how the organism works holistically; such efforts could be the foundation for a new era of personalized medicine. Before long the idea of being treated in a generic way will seem positively medieval; everything, from the kind of care we receive to the medicines we are offered, will be precisely tailored to our DNA and specific biomarkers. Eventually, it might be possible to reconfigure ourselves to enhance our immune responses. That, in turn, might open the door to even more ambitious experimentation like longevity and regenerative technologies, already a burgeoning area of research.

Altos Labs, which has raised $3 billion, more start-up funding than for any previous biotech venture, is one company seeking to find effective anti-

aging technologies. Its chief scientist, Richard Klausner, argues, “We think we can turn back the clock” on human mortality. Focusing on techniques of “rejuvenation programming,” the company aims to reset the epigenome, chemical marks on DNA that control genes by turning them “on” and “off.” As we get older, these “flip” to wrong positions. This experimental approach aims to flip them back, reversing or arresting the aging process. Alongside a host of other promising interventions, the inevitability of physical aging—what seems like a fundamental part of human life—is called into question. A world where life spans are set to average a hundred years or more is achievable in the next decades. Nor is this just about longer life; it’s about healthier lives as we get older.

Success would have major societal repercussions. At the same time, cognitive, aesthetic, physical, and performance-related enhancements are also plausible and would be as disruptive and reviled as they are desired. Either way, serious physical self-modifications are going to happen. Initial work suggests memory can be improved and muscle strength enhanced. It won’t be long before “gene doping” becomes a live issue in sports, education, and professional life. Laws governing clinical trials and experiments hit a gray area when it comes to self-administration. Experimenting on others is clearly off-limits, but experimenting on yourself? As with many other elements of frontier technologies, it’s a legally and morally ill-defined space.

Already the first children with edited genomes have been born in China after a rogue professor embarked on a series of live experiments with young couples, eventually leading, in 2018, to the birth of twins, known as Lulu and Nana, with edited genomes. His work shocked the scientific community, breaching all ethical norms. None of the usual safeguards or accountability mechanisms were in place; the editing was viewed as medically unnecessary and, worse, badly executed. The outrage felt by scientists was real, the condemnation near universal. Calls for a moratorium were swift and included many of the field’s key pioneers, but still, not everyone agreed this was the right approach. Before more CRISPR babies

are born, the world will likely need to grapple with iterated embryo selection that could also select for desired traits.

Apart from the worrying biotech headlines, more and more applications will emerge, a vast array beyond medicine or personal alteration, limited only by the imagination. Manufacturing processes, agriculture, materials, energy generation, even computers—all will be fundamentally transformed in decades to come. While numerous challenges remain, materials core to the economy like plastics, cement, and fertilizer could be produced much more sustainably, with biofuels and bioplastics replacing carbon-emitting incumbents. Crops could become resistant to infection, using less water, land, and fertilizer; houses sculpted and grown from fungi.

Scientists like the Nobel laureate Frances Arnold create enzymes that produce novel chemical reactions, including ways to bind silicon and carbon, usually a tricky, energy-intensive process with wide-ranging uses in areas like electronics. Arnold's method is fifteen times as energy efficient as standard industrial alternatives. The next step involves scaling up production of biological materials and processes. In this way significant products like meat replacements or new materials sucking carbon out of the atmosphere could be grown as much as made. The vast petrochemical industry could see a challenge from young start-ups like Solugen, whose Bioforge is an attempt to build a carbon-negative factory; it would produce a wide range of chemicals and commodities, from cleaning products to food additives to concrete, all while pulling carbon out of the atmosphere. Their process is essentially low-energy, low-waste bio-manufacturing at industrial scale, built on AI and biotech.

Another company, LanzaTech, harnesses genetically modified bacteria to convert waste CO_2 from steel mill production into widely used industrial chemicals. This kind of synthetic biology is helping to build a more sustainable "circular" economy. Next-generation DNA printers will produce DNA with an increasing degree of precision. If improvements can be made in not only expressing that DNA but then using it to genetically engineer a diverse array of new organisms, automating and scaling the processes, a device or set of devices could, theoretically, produce an enormous range of

biological materials and constructions using only a few basic inputs. Want to make some washing detergent or a new toy or even grow a house? Just download the "recipe" and hit "go." In the words of Elliot Hershberg, "What if we could grow what we wanted locally? What if our supply chain was just biology?"

Eventually, computers might also be grown as well as made. Remember that DNA is itself the most efficient data storage mechanism we know of—capable of storing data at millions of times the density of current computational techniques with near-perfect fidelity and stability. Theoretically, the entirety of the world's data might be stored in just one kilogram of DNA. A biological version of a transistor called a transcriptor uses DNA and RNA molecules to act as logic gates. There is still a long way to go before this technology can be harnessed. But all the functional parts of a computer—data storage, information transmission, and a basic system of logic—can in principle be replicated using biological materials.

Already genetically engineered organisms account for 2 percent of the U.S. economy through agricultural and pharmaceutical uses. This is just the start. McKinsey estimates that up to 60 percent of physical inputs into the economy could ultimately be subject to "bio-innovation." Forty-five percent of the global disease burden could be met with "science that is conceivable today." As the tool kit gets cheaper and more advanced, a universe of possibility becomes subject to exploration.

AI IN THE AGE OF SYNTHETIC LIFE

Proteins are the building blocks of life. Your muscles and blood, hormones and hair, indeed, 75 percent of your dry body weight: all proteins. They are everywhere, coming in every conceivable form, doing myriad vital tasks, from the cords holding your bones together, to the hooks on antibodies used to catch unwanted visitors. Understand proteins, and you've taken a giant leap forward in understanding—and mastering—biology.

But there's a problem. Simply knowing the DNA sequence isn't enough to know how a protein works. Instead, you need to understand how it folds. Its shape, formed by this knotted folding, is core to its function: collagen in our tendons has a rope-like structure, while enzymes have pockets to hold the molecules they act on. And yet, in advance, there was no means of knowing how this would happen. If you used traditional brute-force computation, which involves systematically trying all the possibilities, it might take longer than the age of the known universe to run through all the possible shapes of a given protein. Finding out how a protein folds was hence an arduous process, holding back the development of everything from drugs to plastic-eating enzymes.

For decades, scientists had been asking if there was a better way. In 1993, they decided to set up a biannual competition—called Critical Assessment for Structure Prediction (CASP)—to see who could crack the protein folding problem. Whoever gave the best predictions of how a protein might fold would win. CASP soon became the benchmark in a ferociously competitive but tight-knit field. Progress was steady, but with no end in sight.

Then, at CASP13 in 2018, held at a palm-fringed resort in Cancún, a rank outsider entrant arrived at the competition, with zero track record, and beat ninety-eight established teams. The winning team was DeepMind's. Called AlphaFold, the project started during a weeklong experimental hackathon in my group at the company back in 2016. It grew to become a landmark moment in computational biology and provides a perfect example of how both AI and biotech are advancing at speed.

While the second-place team, the well-regarded Zhang group, could predict only three protein structures out of forty-three of the most difficult targets, our winning entry predicted twenty-five. It did this much faster than its rivals, in only a matter of hours. Somehow in this established competition, populated by ultrasmart professionals, our wild card had triumphed and stunned everyone. Mohammed AlQuraishi, a well-known researcher in the field, was left asking, "What just happened?"

Our team used deep generative neural networks to predict how the proteins might fold based on their DNA, training on a set of known proteins and extrapolating from there. The new models were better able to guess the distance and angles of pairs of amino acids. It wasn't expertise in pharma, or in the traditional techniques like cryo-electron microscopy, or even conventional algorithmic methods, that cracked the problem. The key was expertise and capability in machine learning, in AI. AI and biology had decisively come together.

Two years later our team was back. One headline said it all: "One of the Biggest Problems in Biology Has Finally Been Solved," wrote *Scientific American.* A previously hidden universe of proteins was revealed at staggering speed. AlphaFold was so good that CASP was, like ImageNet, retired. For half a century protein folding had been one of science's grand challenges, and then, all of a sudden, it was ticked off the list.

In 2022, AlphaFold2 was opened up for public use. The result has been an explosion of the world's most advanced machine learning tools, deployed in both fundamental and applied biological research: an "earthquake," in the words of one researcher. More than a million researchers accessed the tool within eighteen months of launch, including virtually all the world's leading biology labs, addressing questions from antibiotic resistance to the treatment of rare diseases to the origins of life itself. Previous experiments had delivered the structure of about 190,000 proteins to the European Bioinformatics Institute's database, about 0.1 percent of known proteins in existence. DeepMind uploaded some 200 million structures in one go, representing almost all known proteins. Whereas once it might have taken researchers weeks or months to determine a protein's shape and function, that process can now begin in a matter of seconds. This is what we mean by exponential change. This is what the coming wave makes possible.

And yet this is only the beginning of a convergence of these two technologies. The bio-revolution is coevolving with advances in AI, and indeed many of the phenomena discussed in this chapter will rely on AI for their realization. Think, then, of two waves crashing together, not a wave

but a superwave. Indeed, from one vantage artificial intelligence and synthetic biology are almost interchangeable. All intelligence to date has come from life. Call them synthetic intelligence and artificial life and they still mean the same thing. Both fields are about re-creating, engineering these utterly foundational and interrelated concepts, two core attributes of humanity; change the view and they become one single project.

Biology's sheer complexity opens up vast troves of data, like all those proteins, almost impossible to parse using traditional techniques. A new generation of tools has quickly become indispensable as a result. Teams are working on products that will generate new DNA sequences using only natural language instructions. Transformer models are learning the language of biology and chemistry, again discovering relationships and significance in long, complex sequences illegible to the human mind. LLMs fine-tuned on biochemical data can generate plausible candidates for new molecules and proteins, DNA and RNA sequences. They predict the structure, function, or reaction properties of compounds in simulation before these are later verified in a laboratory. The space of applications and the speed at which they can be explored is only accelerating.

Some scientists are beginning to investigate ways to plug human minds directly into computer systems. In 2019, electrodes surgically implanted in the brain let a fully paralyzed man with late-stage ALS spell out the words "I love my cool son." Companies like Neuralink are working on brain interfacing technology that promises to connect us directly with machines. In 2021 the company inserted three thousand filament-like electrodes, thinner than a human hair, that monitor neuron activity, into a pig's brain. Soon they hope to begin human trials of their N1 brain implant, while another company, Synchron, has already started human trials in Australia. Scientists at a start-up called Cortical Labs have even grown a kind of brain in a vat (a bunch of neurons grown in vitro) and taught it to play *Pong.* It likely won't be too long before neural "laces" made from carbon nanotubes plug us directly into the digital world.

What happens when a human mind has instantaneous access to computation and information on the scale of the internet and the cloud? It's

almost impossible to imagine, but researchers are already in the early days of making it happen. As the central general-purpose technologies of the coming wave, AI and synthetic biology are already entangled, a spiraling feedback loop boosting each other. While the pandemic gave biotech a massive awareness boost, the full impact—possibilities and risks alike—of synthetic biology has barely begun to sink into the popular imagination.

Welcome to the age of biomachines and biocomputers, where strands of DNA perform calculations and artificial cells are put to work. Where machines come alive. Welcome to the age of synthetic life.